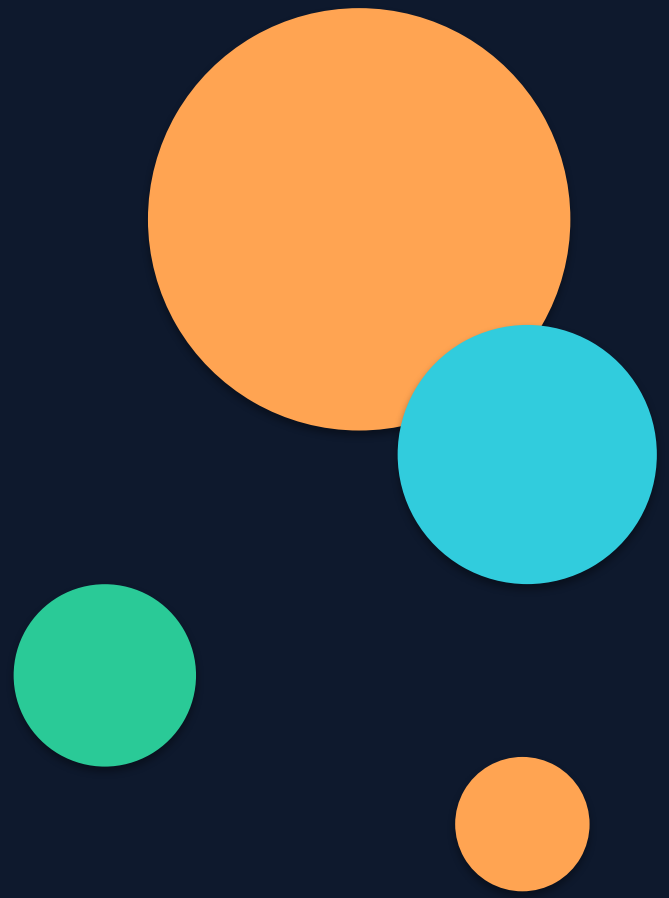
Statistics & Machine Learning



Detailed compact notes with concepts, methods, examples and practical reminders.

6 topic slides + cover = 7 slides

Descriptive Statistics and Probability

Key concepts • practical context • quick revision

- Descriptive statistics summarize the observed sample; inferential statistics estimate or test population claims.
- Mean, median and mode describe central tendency; variance, standard deviation and IQR describe spread.
- Probability ranges from 0 to 1; complementary event probability is 1 − P(A).
- Conditional probability describes P(A given B); independence means one event does not change the other's probability.
- Sampling methods and sample size affect how well results generalize to a population.
- Selection bias can invalidate conclusions even when the sample is large.

EXAMPLE / REMEMBER

Use a representative sampling method before relying on statistical conclusions.

Distributions and Statistical Inference

Key concepts • practical context • quick revision

- Normal distributions are symmetric and described by mean and standard deviation.
- Binomial models count successes across a fixed number of independent trials with constant success probability.
- Sampling distributions describe how an estimator varies across repeated samples.
- Confidence intervals provide a range of plausible values under the method's assumptions.
- Hypothesis tests compare data with a null model; interpret p-values with effect size and context.
- Statistical significance does not automatically imply practical importance or causation.

EXAMPLE / REMEMBER

Report estimate, uncertainty, sample size and assumptions—not only a p-value.

Machine Learning Problem Types

Key concepts • practical context • quick revision

- Supervised learning trains from labeled examples: regression predicts numeric values; classification predicts categories.
- Unsupervised learning finds structure without target labels, such as clusters or reduced representations.
- Reinforcement learning learns a policy through actions, feedback and rewards.
- Common applications: price forecasting, spam detection, customer segmentation and anomaly detection.
- Choose a model based on target, data size, interpretability, latency and error costs.
- Start with a simple baseline and compare more complex methods against it.

EXAMPLE / REMEMBER

Regression: predict a number.
Classification: predict a class.
Clustering: discover groups.

Training, Validation and Overfitting

Key concepts • practical context • quick revision

- Training data estimates model parameters; validation data supports tuning; test data estimates final generalization.
- Overfitting occurs when a model learns noise or specifics of training examples.
- Underfitting occurs when the model is too simple to capture useful patterns.
- Cross-validation estimates performance across multiple training/validation splits.
- Regularization, simpler models, more representative data and careful feature selection can reduce overfitting.
- Keep preprocessing inside a pipeline to avoid leakage across data splits.

EXAMPLE / REMEMBER

Never use the final test set repeatedly to choose model settings.

Evaluation Metrics

Key concepts • practical context • quick revision

- Regression: MAE is average absolute error; RMSE penalizes large errors more strongly.
- Classification: accuracy is correct predictions divided by all predictions.
- Precision = true positives / predicted positives; recall = true positives / actual positives.
- F1 combines precision and recall; a confusion matrix shows types of classification errors.
- With imbalanced classes, accuracy may look high while minority-class detection is poor.
- Choose metrics based on the cost of false positives, false negatives and prediction errors.

EXAMPLE / REMEMBER

For disease screening, missing a true case may cost more than producing a false alarm.

Model Selection and Responsible Use

Key concepts • practical context • quick revision

- Compare against a baseline and use validation results to choose models.
- Inspect performance across relevant subgroups to detect uneven errors or bias.
- Check data quality, target definition and representativeness before interpreting a score.
- Explain limitations, uncertainty and the intended scope of use to stakeholders.
- Use human review where errors can have significant consequences.
- Monitor real-world performance because data and user behavior can change.

EXAMPLE / REMEMBER

Best model means best for the real objective and constraints—not necessarily the most complex.